Can an LLM Agent's "Done" Be Trusted? Scoring a Desktop Agent's Success Reports Against World-State Ground Truth

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Abstract. LLM agents that act on a computer end each task with a report to the user, "done" or "failed". I treat these reports as a binary classifier and score them against world-state ground truth on a 44-task protocol fixed before the run (31 trials scored) for BeepSME, a publicly released voice-first desktop agent. The agent's verdict was right on 21 of 31 trials (accuracy 0.677, 95% CI 0.501 to 0.814) and erred equally in both directions: 5 false "done" and 5 false "failed". A post-hoc self-confidence score ranked successes above failures (ROC AUC 0.798, bootstrap 95% CI 0.677 to 0.906). No decision rule examined kept both error types low. A blind human second rating agreed with the first (LLM) rater on 6 of 8 trials (Cohen's kappa 0.47). The report also documents a logging defect that had made an earlier result invalid.

1 Motivation

When an agent acts for a person, its end-of-task report is often the only feedback that person receives. If the report says "done" when the task failed (over-confidence), the error is passed on silently; if it says "failed" when the task succeeded (over-deferral), the person repeats work or loses trust. I ask how often each happens for a deployed agent, and how the choice of verification rule moves errors between the two.

2 Method

Tasks. 44 tasks over 7 skills (teach, homework, read aloud, translate, write, tutor, send), each with an oracle written before the run and stratified into happy-path, over-confidence bait, over-deferral bait and impossible tasks. The protocol was committed to version control on 2026-06-05 and run on 2026-08-20. 31 trials were scored; 12 were not run (WhatsApp send and Canva tutor tasks needed accounts) and 1 was excluded.

Ground truth. Each trial was labelled 1 or 0 from world state (saved files, answer keys, captured speech, the screen), with a separate label for whether the task was achievable at all. The first rater was an LLM working from a sheet that hid the agent's report and confidence. I then labelled a seeded random sample of ten trials blind to the agent's report, its confidence and the first rater's labels.

Predictors. (a) The agent's end-of-task verdict, recovered from its run logs. (b) The research-log field the protocol named as the claim. (c) A 0-100 self-confidence score from one extra grader call per task. Field (b) turned out to be hardcoded to "success" for any turn that did not crash, so the primary predictor is (a); that choice was made after seeing the data and is reported as a deviation.

3 Results

Decision rule	Reports success on	False "done" (FP)	False "failed" (FN)
Logged claim field	31	14	0
Agent's verdict	17	5	5
Self-confidence ≥ 35	5	0	12

Table 1. Errors of three decision rules on the same 31 trials (17 successes, 14 failures).

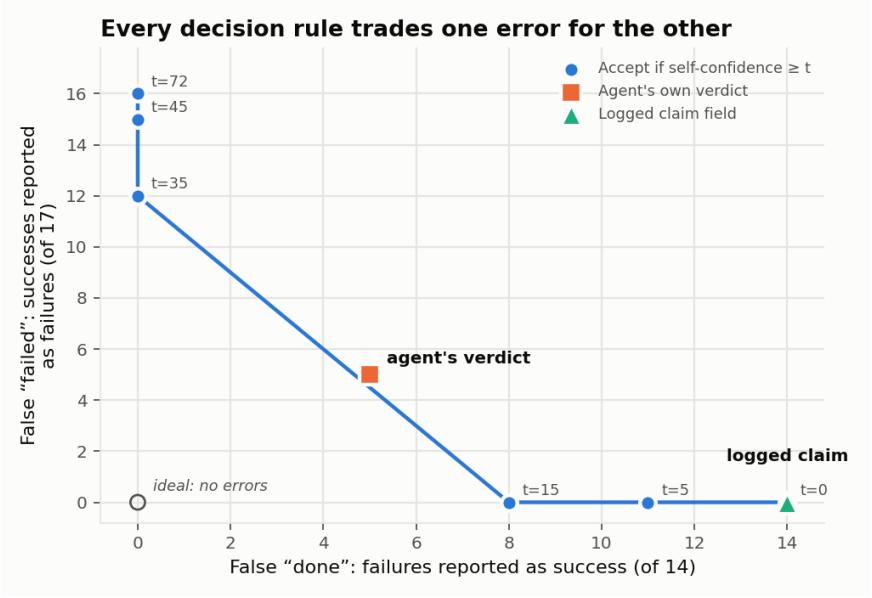


Figure 1. Every rule trades one error type for the other; none reaches the no-error corner.

The verdict's errors have identifiable causes. All 5 false "failed" reports share one cause (ai_http_413, a request-size error raised after the content had been delivered). 4 of the 5 false "done" reports were tasks that could not be done: the verdict has no "declined" outcome, so an honest refusal and a silent no-op both end as success. The self-confidence score was informative but coarse: mean 23.65 on successes and 9.64 on failures, with no failure above 15, but 20 of 31 trials at exactly 15.

4 Agreement between raters

On the ten re-labelled trials, 2 could not be decided from the published evidence. On the other 8, the raters agreed on 6 (observed agreement 0.75, kappa 0.47). One disagreement was whether a poem spoken aloud satisfies a request for a document: the raters differed on what success means, a question about the type and format of feedback rather than about what happened.

5 Limitations

A small sample from one agent on one day; wide intervals. The first rater is from the same model family as the agent, and the human rater built the system and knew the headline results. Where declining is the right behaviour, the world-state label is still 0. The confidence grader was always told the outcome was "success". Screenshots and full logs stay private; the repository documents how the published data were extracted from them.

6 Next steps

Add a "declined" outcome and score refusals separately; tie the verdict to what the user received; combine self-reports, LLM judges and sparse human labels while learning each source's reliability, in the spirit of hybrid human-AI evaluation (Mesbah et al., 2023); and test in user studies which report format helps people catch an agent's mistakes.

References

Mesbah, S., Arous, I., Yang, J., and Bozzon, A. (2023). HybridEval: A Human-AI Collaborative Approach for Evaluating Design Ideas at Scale. The Web Conference (WWW).

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